

Abstract

Semantic Enrichment of Nigerian Pidgin English for Contextual Sentiment Classification Nigerian English adaptation, Pidgin, has evolved over the years through multi-language code switching, code mixing and linguistic adaptation. While Pidgin preserves many of the words in the normal English language corpus, both in spelling and pronunciation, the fundamental meaning of these words have changed significantly. For example, 'ginger' is not a plant but an expression of motivation and 'tank' is not a container but an expression of gratitude. The implication is that the current approach of using direct English sentiment analysis of social media text from Nigeria is sub-optimal, as it will not be able to capture the semantic variation and contextual evolution in the

contemporary meaning of these words. In practice, while many words in Nigerian Pidgin adaptation are the same as the standard English, the full English language based sentiment analysis models are not designed to capture the full intent of the Nigerian pidgin when used alone or code-mixed. By augmenting scarce human labelled code-changed text with ample synthetic code-reformatted text and meaning, we achieve significant improvements in sentiment scoring. Our research explores how to understand sentiment in an intrasentential code mixing and switching context where there has been significant word localization. This work presents a 300 VADER lexicon compatible Nigerian Pidgin sentiment tokens and their scores and a 14,000 gold standard Nigerian Pidgin tweets and their sentiments labels.

Background

Language is evolving with the flattening world order and the pervasiveness of the social media in fusing culture and bridging relationships at a click. One of the consequences of the conversational evolution is the intrasentential code switching, a language alternation in a single discourse between two languages, where the switching occurs within a sentence BIBREF0. The increased instances of these often lead to changes in the lexical and grammatical context of the language, which are largely motivated by situational and stylistic factors BIBREF1. In addition, the need to communicate effectively to different social classes have further orchestrated this shift in language meaning over a long period of time to serve socio-linguistic functions BIBREF2 Nigeria is estimated to have between three and five million people, who primarily use Pidgin in their day-to-day interactions. But it is said to be a second language to a much higher number of up to 75 million people in Nigeria alone, about half the population.BIBREF3. It has evolved in meaning compared to Standard English due to intertextuality, the shaping of a text's meaning by another text based on the interconnection and

influence of the audience's interpretation of a text. One of the biggest social catalysts is the emerging urban youth subculture and the new growing semi-literate lower class in a chaotic medley of a converging megacity BIBREF4 BIBREF5 VADER (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon and rule-based sentiment analysis tool that is specifically attuned to sentiments expressed in social media and works well on texts from other domains. VADER lexicon has about 9000 tokens (built from existing well-established sentiment word-banks (LIWC, ANEW, and GI) incorporated with a full list of Western-style emoticons, sentiment-related acronyms and initialisms (e.g., LOL and WTF)commonly used slang with sentiment value (e.g., nah, meh and giggly)) with their mean sentiment rating.BIBREF6. Sentiment analysis in code-mixed text has been established in literature both at word and sub-word levels BIBREF7 BIBREF8 BIBREF9. The possibility of improving sentiment detection via label transfer from monolingual to synthetic code-switched text has been well executed with significant improvements in sentiment labelling accuracy (1.5%, 5.11%, 7.20%) for three different language pairs BIBREF5

Method

This study uses the original and updated VADER (Valence Aware Dictionary and Sentiment Reasoner) to calculate the compound sentiment scores for about 14,000 Nigerian Pidgin tweets. The updated VADER lexicon

(updated with 300 Pidgin tokens and their sentiment scores) performed better than the original VADER lexicon. The labelled sentiments from the updated VADER were then compared with sentiment labels by expert Pidgin English speakers.

Results

During the translation of VADER English lexicon to suitable one-word Nigerian Pidgin translation, a total of 300 Nigerian pidgin tokens were successfully translated from the standard VADER English lexicon. One of the

challenges of this translation is that the direct translation of most the sentiment words in the original VADER English Lexicon translates to phrases not single one-word tokens and certain pidgin words translates to many english words.TABREF5.

Conclusion

The quality of sentiment labels generated by our updated VADER lexicon is better compared to the labels generated by the original VADER English lexicon.

Human annotators were able to capture nuances that the rule based sentiment labelling could not capture. More work can be done to increase the number of instances in the dataset.

Appendix ::: Selection of Data Labellers

Three people who are indigenes or lived in the South South part of Nigeria, where Nigerian Pidgin is a prevalent

method of communication were briefed on the fundamentals of word sentiments. Each labelled Data point was verified by at least one other person after initial labelling.

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